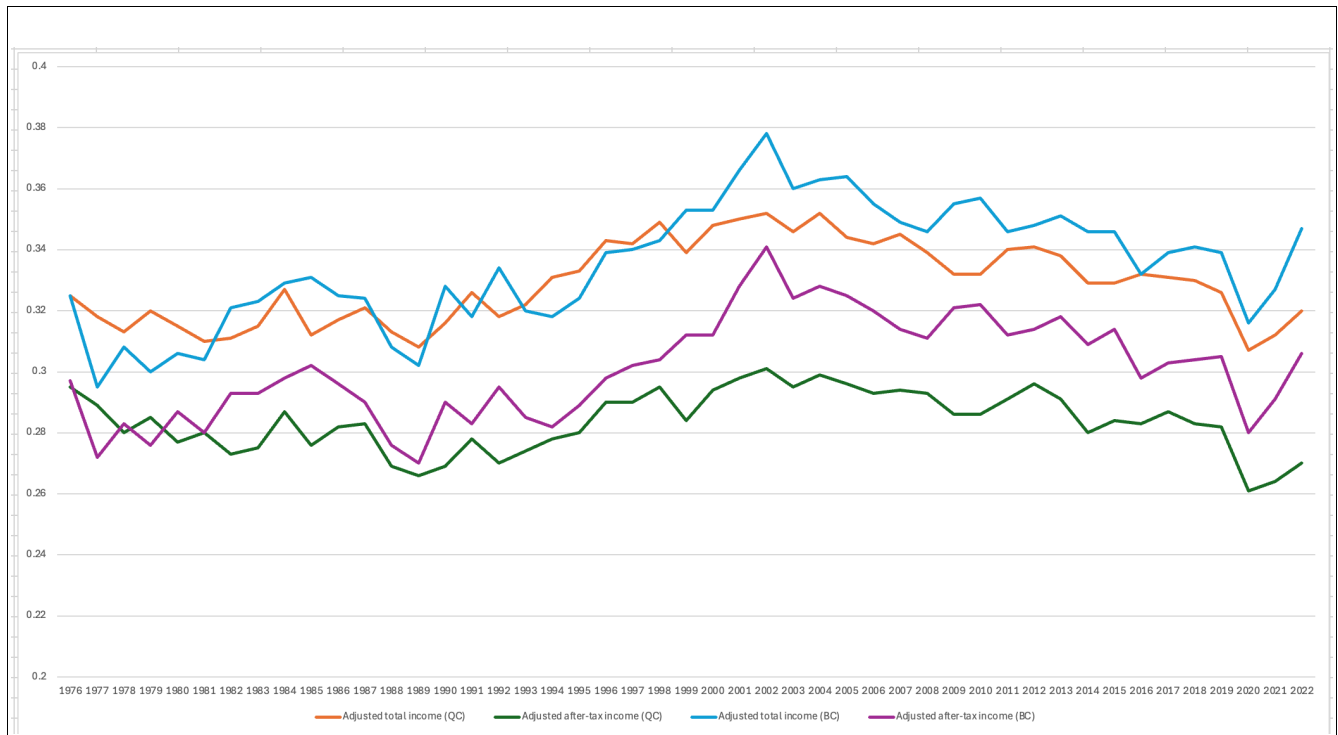


SUBMIT AS A PDF!

At home component of final exam Access Statistics Canada's table 11-10-0134-01 and download the Gini coefficients for total income and after tax income for Quebec and British Columbia from the earliest to the most current date. Answer the following questions in the space provided:

a) Graph these Gini measures for Quebec (QC) and British Columbia (BC) across time. You should have 4 lines graphed. Make sure that you adjust the Y-axis so that it starts at 0.2 (otherwise the lines will be too squished together) (2)



b) Describe 3 things you observe when comparing 'Total' and 'after-tax' Gini trends for QC & BC. (3)

After-tax Gini is consistently lower than total Gini for both QC and BC. In both provinces, the Gini based on after-tax income (green line for BC, purple line for QC) always sits below the Gini based on total income (blue line for BC, orange line for QC). This gap reflects the impact of taxes and transfers in reducing income inequality. While the exact size of the gap varies by year, after-tax income distributions are invariably more equal than total income distributions.

BC's Gini (both total and after-tax) tends to be higher than QC's. Throughout most of the timeline, British Columbia's total Gini (blue) is higher than Quebec's total Gini (orange), and BC's after-tax Gini (green) is likewise above QC's after-tax Gini (purple). This means, BC exhibits higher inequality before and after taxes compared to QC. The difference between the two provinces is typically more pronounced in total income, suggesting that labor-market or pre-tax factors in BC contribute to higher inequality.

Both provinces show parallel ups and downs over time, but BC exhibits somewhat larger swings. From the early 1980s through the 2000s, both BC and QC display periods of rising Gini followed by dips noticeable in the mid-1980s, mid 1990s, and around the mid 2000s. These ups and downs appear in both total and after-tax measures. However, BC's total Gini (blue) tends to peak slightly higher (e.g., around the mid 2000s), and its after-tax Gini (green) also shows larger variations relative to QC's. That wider fluctuation suggests that BC's inequality (both before and after taxes) can be more sensitive to economic cycles or changes in redistribution policy than QC's.